

DRAFT

Notes and User Manual for the VACUUM Code

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These notes are complementary to the vacuum paper, “Vacuum Calculations in Azimuthally Symmetric Geometry”, M. S. Chance, *Phys. Plasmas* (4), June, 1997, 2161-2180.

1 Preliminaries

Because of historical reasons and the continual addition of features and refinements, the look and feel of the data may not seem polished and streamlined. It will someday. Thus, some variables are redundant and some of the control variables [logical and integers] may seem to overlap. In general, it's best to turn off switches which are not relevant to your case.

The code has been used most extensively for up-down symmetric configurations and so most confidence in the results are for this symmetry. It has been used for simple applications for non up-down symmetric configurations but some of the ancillary applications (like those in the diagnostic capabilities) have not been thoroughly checked out as of now.

- Radius means along the horizontal.
- The following are excerpted from the published paper. The latter contains more details about the various shell options and the Mirnov loop and eddy current diagnostic capabilities.
 - Some definitions used here are the plasma ‘radius’, p_{rad} and the plasma center $(X_{\text{cen}}, Z_{\text{cen}})$, which are defined by:

$$p_{\text{rad}} \equiv \frac{1}{2}(X_{\text{max}} - X_{\text{min}}) \quad (1)$$

$$X_{\text{cen}} = \frac{1}{2}(X_{\text{max}} + X_{\text{min}}) \quad (2)$$

and

$$Z_{\text{cen}} = \frac{1}{2}(Z_{\text{min}} + Z_{\text{max}}), \quad (3)$$

where X_{min} and X_{max} are respectively, the minimum and the maximum X -values of the plasma, etc.

- There is an option in the code which numerically redistributes the grid along the shell so that the grid is equidistant between adjacent points. While this usually gives the best overall accuracy, this option should not be exercised when the wall is very close to the plasma since the plasma-wall points may not be aligned with respect the indexing. In this case, it is best to choose the equidistant-shell option, (`ishape` = 6), with the equal-arc option for the shell turned off, (`leqarcw` = 0).
- The definitions of *elongation* and *triangularity* can be ascertained from the following description of the **Simple toroidal D-shaped shell**:

This configuration where the shell is centered at c_w , has an elongation factor of b_w , a triangularity skewness d_w , and a radius in the equatorial plane of a , is given by the relation,

$$X_{w_i} = c_w + a \cos(\theta_i + d_w \sin \theta_i) \quad (4)$$

$$Z_{w_i} = -b_w a \sin \theta_i. \quad (5)$$

This gives a free-standing shell independent of the plasma position or dimensions, and is useful if the shell is from the simulation of an actual device.

- An example of the parameterization of the shell which is scaled to the plasma is given by,

$$X_{w_i} = X_{\text{cen}} + p_{\text{rad}} c_w + p_{\text{rad}} (1 + a - c_w) \cos(\theta_i + d_w \sin \theta_i) \quad (6)$$

$$Z_{w_i} = -b_w p_{\text{rad}} (1 + a - c_w) \sin \theta_i. \quad (7)$$

Here the center of the shell is offset by $p_{\text{rad}} c_w$ in the equatorial plane and is still a distance $p_{\text{rad}} a$ from the outer major radius side of the plasma.

2 Notes on the Normalization of δW in VACUUM

The VACUUM code calculates the response to

$$\mathcal{B} = \mathcal{J} \nabla \chi \cdot \nabla \mathcal{Z}, \quad (8)$$

with

$$\mathcal{J} = (\nabla \mathcal{Z} \cdot \nabla \theta \times \nabla \phi)^{-1}, \quad (9)$$

$\mathcal{Z}$ the radial coordinate and χ the magnetic scalar potential.

At the interface

$$\mathcal{B} = \mathcal{J} \mathbf{Q} \cdot \nabla \mathcal{Z} \quad (10)$$

$$= \mathcal{J} \nabla \mathcal{Z} \cdot \nabla \times (\boldsymbol{\xi} \times \mathbf{B}) \quad (11)$$

$$= \mathcal{J} \mathbf{B} \cdot \nabla (\boldsymbol{\xi} \cdot \nabla \mathcal{Z}). \quad (12)$$

Now the equilibrium field is

$$\mathbf{B} = \nabla \phi \times \nabla \psi + R g \nabla \phi \quad (13)$$

$$= \psi' \nabla \phi \times \nabla \mathcal{Z} + R g \nabla \phi, \quad (14)$$

where $\psi = \Psi/2\pi$, Ψ being the poloidal flux. So

$$\mathbf{B} \cdot \nabla \theta = \mathcal{J}^{-1} \psi' \quad (15)$$

where the prime denotes derivative with respect to $\mathcal{Z}$. Then we have

$$\mathcal{B} = \psi' (m - nq) \boldsymbol{\xi} \cdot \nabla \mathcal{Z} \quad (16)$$

in a straight-field-line coordinate system.

In the cylinder with r as the radial coordinate,

$$\psi' = R B_\theta \quad (17)$$

$$= \frac{r B_z}{q}, \quad (18)$$

whereas if the radial coordinate is chosen to be ψ , then $\psi' = 1$, and the ratio between these two cases is $r B_z/q$. This is equal to $1/q$ if $r = 1$ and B_z is unity.

It should be noted that VACUUM calculates the response to $\mathcal{B}$ but oftentimes outputs the vacuum matrix $V_{mm'}$ as the response to $\boldsymbol{\xi}$, in which case the factor of $(m - nq)(m' - nq)$ arising from Eqn. 16 is incorporated in $V_{mm'}$, where $m = m'$ in the circular cylinder.

Further normalizations for interfacing to specific codes may be necessary for compatibility. See for example, Sec. 5.

3 Form of the Source Term

The magnetic field $\nabla\chi(\theta, \zeta)$ in the vacuum region is driven by the normal component of the perturbed magnetic field at the plasma-vacuum interface. It naturally enters the equations in the VACUUM code in the form

$$\mathcal{B}(\theta, \zeta) = \mathcal{J} \nabla\chi \cdot \nabla\mathcal{Z}, \quad (19)$$

$$= X \left(X_\theta \frac{\partial\chi}{\partial Z} - Z_\theta \frac{\partial\chi}{\partial X} \right), \quad (20)$$

so that, given the parametrization, $[X(\theta), Z(\theta)]$, $\mathcal{B}$ doesn't involve the Jacobian directly and the Neumann problem is independent of the coordinate used.

If Q_n is given, then $\mathcal{B} = \mathcal{J} |\nabla\mathcal{Z}| Q_n = X [X_\theta^2 + Z_\theta^2]^{1/2} Q_n$.

Fourier analysis can be carried out in any (θ, ζ) system:

$$\mathcal{B}(\theta, \zeta) = \sum_l \mathcal{B}_l e^{i(l\theta - n\zeta)} \quad (21)$$

and χ is calculated as a response matrix to the vector containing $\mathcal{B}_l$. Note that the two dimensional Green's function used by VACUUM requires a $\exp(-in\phi)$ behavior, so the deviation from ϕ must be taken into account, like e.g., in the **straight field lines** case below.

If $\xi_{\mathcal{Z}}(\theta, \zeta) \equiv \xi \cdot \nabla\mathcal{Z}$ is supplied as the driving term, then in any θ ,

$$\mathcal{B}(\theta, \zeta) = \mathcal{J} \mathbf{Q} \cdot \nabla\mathcal{Z} \quad (22)$$

$$= \mathcal{J} \mathbf{B} \cdot \nabla \xi_{\mathcal{Z}}(\theta, \zeta) \quad (23)$$

$$= \mathcal{J} \mathbf{B} \cdot \nabla\theta \left[\frac{\partial}{\partial\theta} - in \frac{\mathbf{B} \cdot \nabla\zeta}{\mathbf{B} \cdot \nabla\theta} \right] \xi_{\mathcal{Z}}(\theta, \zeta), \quad (24)$$

so information about both $\mathbf{B}$ and $\mathcal{J}$ is needed.

3.1 Straight field lines

If the coordinates are such that the field lines are straight then the relation,

$$\mathcal{B}_l \sim i(l - nq)\xi_l, \quad (25)$$

holds, and then the θ must be chosen to be the PEST angle, θ_p if the toroidal angle is ϕ .

Alternatively, for an arbitrary choice of θ , we require the parametrization of $X(\theta)$, $Z(\theta)$, and the coordinate transformation contained in $\nu(\theta)$, where $\zeta = \phi + \nu(\theta)$. Then the Fourier coefficients of $\mathcal{B}$ can be written as Eq. (25).

4 The χ Response and the Vacuum Energy

The code calculates the response, $\mathcal{C}$, of χ to the plasma perturbation, $\mathcal{B}$, so that,

$$\chi_i(\theta_i) = \mathcal{C}_{ik}(\theta_i, \theta'_k) \mathcal{B}(\theta'_k). \quad (26)$$

Inserting the $\exp(-in\phi)$ dependence explicitly and separating the real and imaginary parts, we have for the response,

$$\chi(\theta) = \oint [\{ (C^R \mathcal{B}^R - C^I \mathcal{B}^I) \cos n\phi + (C^R \mathcal{B}^I + C^I \mathcal{B}^R) \sin n\phi \} \quad (27)$$

$$+ i \{ (C^R \mathcal{B}^I + C^I \mathcal{B}^R) \cos n\phi - (C^R \mathcal{B}^R - C^I \mathcal{B}^I) \sin n\phi \}] d\theta'. \quad (28)$$

In the Fourier basis given by Eq. (21), we find,

$$\chi(\theta) = e^{-in\phi} \sum_l [X_l^R(\theta) + i X_l^I(\theta)] \quad (29)$$

or

$$\chi(\theta) = \sum_l [X_l^R(\theta) \cos n\phi + X_l^I(\theta) \sin n\phi + i \{X_l^I(\theta) \cos n\phi - X_l^R(\theta) \sin n\phi\}]; \quad (30)$$

one also needs

$$\frac{\partial \chi}{\partial \phi} = -n \sum_l [X_l^R(\theta) \sin n\phi - X_l^I(\theta) \cos n\phi + i \{X_l^I(\theta) \sin n\phi + X_l^R(\theta) \cos n\phi\}]. \quad (31)$$

Here

$$X_l^R(\theta) = \mathcal{B}_l^R \mathcal{C}_l^R(\theta) - \mathcal{B}_l^I \mathcal{C}_l^I(\theta), \quad (32)$$

$$X_l^I(\theta) = \mathcal{B}_l^R \mathcal{C}_l^I(\theta) + \mathcal{B}_l^I \mathcal{C}_l^R(\theta), \quad (33)$$

with

$$\mathcal{C}_l^R(\theta) = \oint \mathcal{C}(\theta, \theta') \cos(l\theta' - n\nu(\theta')) d\theta', \quad (34)$$

$$\mathcal{C}_l^I(\theta) = \oint \mathcal{C}(\theta, \theta') \sin(l\theta' - n\nu(\theta')) d\theta'. \quad (35)$$

The vacuum energy can be written as,

$$2\delta W_v = 2\pi \sum_{ik}^p \mathcal{C}_{ik} \mathcal{B}^*(\theta_k) \mathcal{B}(\theta_i), \quad (36)$$

where the notation $\sum^p$ denotes that the summation involves only the points on the plasma surface. The factor of 2π arose from the integral over ϕ . This form should hold regardless of whether the basis is Fourier or finite elements. More explicitly, we have,

$$2\delta W_v = 2\pi \sum_{kk'}^p [(\mathcal{B}_k^R \mathcal{C}_{kk'}^R \mathcal{B}_{k'}^R - \mathcal{B}_k^I \mathcal{C}_{kk'}^I \mathcal{B}_{k'}^I) + (\mathcal{B}_k^I \mathcal{C}_{kk'}^R \mathcal{B}_{k'}^I + \mathcal{B}_k^R \mathcal{C}_{kk'}^I \mathcal{B}_{k'}^I) \quad (37)$$

$$+ i \{(\mathcal{B}_k^I \mathcal{C}_{kk'}^R \mathcal{B}_{k'}^R + \mathcal{B}_k^R \mathcal{C}_{kk'}^I \mathcal{B}_{k'}^I) - (\mathcal{B}_k^R \mathcal{C}_{kk'}^R \mathcal{B}_{k'}^I - \mathcal{B}_k^I \mathcal{C}_{kk'}^I \mathcal{B}_{k'}^I)\}]. \quad (38)$$

If the matrix $\mathcal{C}_{kk'}$ is symmetric, cancellations reduce this to

$$2\delta W_v = 2\pi \sum_{kk'}^p [(\mathcal{B}_k^R \mathcal{C}_{kk'}^R \mathcal{B}_{k'}^R + \mathcal{B}_k^I \mathcal{C}_{kk'}^R \mathcal{B}_{k'}^I) \quad (39)$$

$$+ i (\mathcal{B}_k^R \mathcal{C}_{kk'}^I \mathcal{B}_{k'}^R + \mathcal{B}_k^I \mathcal{C}_{kk'}^I \mathcal{B}_{k'}^I)]. \quad (40)$$

5 GATO specifics

The control variables and values that are specific for interfacing VACUUM to GATO are `lrgato = 2` and `lgato = [0, 1, 2, 3]`. Be sure that the variables `ladj`, `ldcon`, `lnova`, `lzio` are set to 0. A model of the DIII-D wall shape is obtained with `ishape = 8`. But note: specifying `ishape = 9` instructs VACUUM to use the variables, `(xwin, zwin)` read in from **vacin** when the option `lrgato = 2` is specified. See the **Input data** section, Sec. 6.1 for a description of these variables. See also `mth`, `mfel`, `lgaus`, `lnint`, `lxint` in Sec. 6.1

As long as `lrgato` is greater than zero, VACUUM will read from **vacin**. The first set of variables in the latter contains information to calculate the coefficient matrix for χ of the source field $\mathcal{B}$. The second set of variables in **vacin** contains the surface eigenfunction of $\mathcal{B}$ needed for the diagnostic section of VACUUM. Since this is unknown until GATO is executed with the boundary information from VACUUM, the first run must be made with the control variable `ieig = 0`. (Unless, of course, GATO is executed with its own, or some other, vacuum calculation). GATO is then executed to obtain the eigenfunction

information. **vacin** is then updated and VACUUM can be executed with the option `ieig = 9` to calculate further diagnostic results. However, for the diagnostic section VACUUM uses the Fourier representation so the code will set `lkdis = 0`, `iloop = 0` if `lgato = 3`. So run with `lgato = 0, 2 lrgato .gt. 0`. Relevant to this, see the variables `lkdis`, `ieig`, `iloop` in Sec. 6.6.

In order to be in agreement with the residing vacuum calculations of GATO the vacuum response matrix of VACUUM is multiplied by a factor of

$$\left(\frac{R_{\text{gato}}}{2\pi M_{\text{fel}}} \right) \left(\frac{\pi}{2\mu_0 R_{\text{gato}}} \right) = \frac{1}{4\mu_0 M_{\text{fel}}},$$

where R_{gato} is passed from GATO through **vacin**, and M_{fel} is the number of finite elements.

6 Input data

The control input file is called **modivmc**. [A variable input file on the command line can be added later]. The names of the input data files needed from the various stability codes and the output files are given in the tables. The notation used corresponds more or less to the coding.

In the first column of the tables is an explicit list of the input variables. The second column shows the equivalent variables used in expressions in the published paper, [*Physics of Plasmas*, **4**, June 1997, 2161]. The third column gives the type of variable and values while the fourth column gives a description of the variables.

The subsections describing the input variables are labeled like the namelist blocks of the input file. Again note that these names are not completely correlated with the kind of variables so the functionality of some of the variables may span the two or more blocks. This will be fixed soon.

6.1 modes

Variable	Paper equiv.	Value or type	Definition
lzio	<i>n</i>	1	Reads from “mp0” and “mp1” for PEST-type inputs.
mp0, mp1		0	Turns off references to “mp0” and “mp1”
mth		File names	Plasma input from PEST-type codes.
		Even integer	Number of grid points used for the calculation. The values of the needed quantities on these points are interpolated from the those gotten from the plasma information in ‘mp0’, ‘mp1’, vacin , invacuum , or equivalent.
lmin, lmax		arrays	Range of Fourier harmonics. Usually the same as that used in the plasma calculation. Not relevant for DCON. In GATO, and for the nonzero lspark options these are used for internal Fourier matrix operations.
lnint, lxint		Integers	Range of the internal Fourier harmonics used for the lspark and lgato = 2 options. Best if lnint = - lxint .
dm0		Integer	Finite element transformation such that elements are centered at $\theta_k = 2\pi(k - 1 + dm0)/N$, $k = 1, N$
dx0, dx1		Integers	Starts new origin of data at $dx1/mth + dx0/mthin$. Interpolates from mthin to mth points. <i>vecin</i> preserved.
mfel		integer	Number of finite elements when the lgato option is turned on.
lgaus			Needed for the lgato = 3 option. The first digit[3, or 5] and second digit[4, or 6] is the order of the Gaussian integration over the observer and source elements respectively.
m			<i>not relevant.</i>
mdiv			<i>not relevant.</i>
n			The toroidal mode number. Overwritten by input from DCON or GATO
lsymz		.true.	Symmetrizes the vacuum matrix
lfunint		logical	<i>not relevant.</i>
xiin[i,r]		array	Input Fourier modes of complex $\xi_l(edge)$. See ieig in Sec. 6.6
leqarcw		1	Turns on equal arcs distribution of the nodes on the shell. Best results unless the wall is very close to the plasma. See ishape = 6 option.
ladj		1	Read input inadjv from and interfaces to the ADJ code. Outputs vacadj
		0	Otherwise
ldcon		1	Reads input from DCON’s invacuum and outputs vacdcon
lgato		> 0	Calculates finite element representation of and outputs the vacuum matrix, vacmat? , for the GATO code.
		1	Direct finite element representation with collocation. Loops and Eddy current not valid. See GATO specifics
		2	Transformation to finite elements from Fourier representation.
		3	Direct finite element representation without collocation. Loops and Eddy current not valid. See GATO specifics

modes, cont'd

Variable	Paper equiv.	Value or type	Definition
lrgato		> 0	Reads file vacin for input for the GATO code.
lspark		> 0	SPARK and feedback type calculations. This is work in progress. Set it to 0.
ismth		1	Smooths some quantities for plotting.
lnova		2	Interfacing to the NOVA code. Takes PEST-type input. Outputs va-cout
lpest1		1	Interfacing to the PEST1 code
		.true.	Interfacing to the PEST2 code (the default) and all other codes but PEST1. PEST-type outputs are in vacmat
lhighn		.false.	Turns on the “high n” transformation to extract constant phase along the magnetic field.
ltnrm		integer	Coefficient of transformation matrix to finite elements.
		1	True norm.
		2	Fake norm: corresponds to the point transformation.

6.2 rshel

Variable	Paper equiv.	Value or type	Definition
lreshel		> 0	Turns on the resistive shell feedback mode.
mshel		Even integer	Intended for grid points on shell. Not yet implemented.
lwmx, lwmn		Integers	Range of the shell shell operator eigenfunctions used in the resistive wall mode, $l_{rshel} > 0$, options.
taugam			The resistive time operator $(\mu\Delta Z/\eta)\gamma$ normalized to the minimum shell eigenvalue.

6.3 debugs

Variable	Paper equiv.	Value or type	Definition
checkd		logical	Switch to: write out matrix elements. Fourier analyze matrix elements and write out the Fourier matrix.
checke		logical	Switch to turn on the calculation of and to write out the eigenvalues and eigenfunctions of selected matrices. Plots χ on the wall.
check1		logical	Turns on timer at selected points in code. It's functional only for the C-90.
check2		logical	Turns on write statements for some quantities for debugging. $\delta(\theta)$, wall coordinates and their derivatives, x_{pass} , z_{pass} , ell , $thgr$, x_{grd} , z_{grd} and usually indices, sums etc.
checks		logical	Switches for checking quantities in the spark module.
lkplt		1	Turns on eddy current plots. Also plots plasma and wall coordinates.
wall		.false.	Leave this .false. [Not used??]

6.4 vacdat

Variable	Paper equiv.	Value or type	Definition
ishape		integer < 0 < 100 Consult the paper for more details: 2 3 4 5 6 7 8 9 < 200 110 120 130 < 300 210 240 < 400 310 340	Options for the wall shape. Spherical topogy. Closed toroidal topology. Elliptical shell confocal to the plasma's radius and height. The radius of the shell is a . Wall of NSTX. Data obtained from J. Menard Modified dee-shaped wall independent of plasma geometry with x-triangularity, d_w , squareness(a_w ?) and 2nd harmonic of z_{wall} in a_w and t_w . Centered at c_w , radius r_w and elongation b_w . dee-shaped wall scaled to the radius and geometric center of the plasma. Offset of c_w . Other variables as option 4. Conforming shell at distance $a * p_{rad}$. This is best option for a close fitting shell since the plasma and shell nodes are aligned. Enclosing bean-shaped wall. Consult paper. Wall of DIII-D. Wall in x_{win} and z_{win} . Read in from, e.g., vacin Solid conductors not linking plasma. dee-shaped conductor. Solid bean-shaped conductor on right. Solid bean-shaped conductor on left. Toroidal conductor with a toroidally symmetric gap. Geometry correlated to plasma position and geometry. shell scaled to plasma. Gap on the inner side. shell scaled to plasma. Gap on the outer side. Toroidal conductor with a toroidally symmetric gap. Geometry independent of plasma. shell independent of plasma. Gap on the inner side. shell independent of plasma. Gap on the outer side.
a_w	a_w		Half-thickness of the shell.
b_w	b_w		Elongation of the shell.
c_w	c_w		Offset of the center of the shell from the major radius, X_{maj} .
d_w	δ_w		Triangularity of shell.
t_w	τ_w		Sharpness of the corners of the shell. Try .05 as a good initial value.
nsing			<i>not relevant</i>
epsq			<i>not relevant</i>
noutv			Number of grid points for the eddy current plots.
delg		noninteger	Size of arrows for the eddy current plots. Integer part is length of shaft and decimal part is size of the head.
idgt		integer	<i>not relevant</i> now. Used to be approx. number of digits accuracy in the Gaussian elimination used in the calculation. A value of $idgt=6$ is usually sufficient.
delfac			Controls grid size to calculate derivatives in spark type calculations.
idsk			<i>not relevant.</i>
cn0			Constant added to the $\mathcal{K}$ matrix to make it nonsingular for $n = 0$ modes.

6.5 shape

[illegible]

6.6 diags

Variable	Paper equiv.	Value or type	Definition
lkdis		integer	Turns on the eddy current calculations. It calls subroutine <code>kdis</code> .
ieig		integer	Options for getting the surface eigenfunctions ξ_l .
		0	Do not read eigenmode information.
		1	From PEST-1. Writes the ω^2 and ξ_l .
		4	Reads from file outdist .
		5	Gets ξ_l from the input <code>xiin</code> in namelist modes.
		6	$\Re[\xi_l]$ and $\Im[\xi_l]$ from input file dcon_ev in subroutine dcnvac for DCON's input.
		8	$\Re[\xi_k]$ and $\Im[\xi_k]$ from input file vacin for GATO's input.
		9	$\Re[Q_k]$ and $\Im[Q_k]$ from input file vacin for GATO's input.
iloop		integer > 0	Turns on Mirnov coils calculation.
		1	Coils location given by dee-shaped geometry set by parameters given below.
		2	PBX's Mirnov coil positions.
		3	Loops at constant distance, <code>aloop*plrad</code> from plasma.
		4	Loops at constant distance, <code>aloop*plrad</code> from shell.
x,zloop			Input positions of Mirnov loops when <code>lpsub = 1</code>
nloop			number of coils around the plasma.
nloopr			number of radial loops.
lpsub		integer	Uses subroutines for coil positions. Otherwise uses namelist inputs (<code>xloop</code> , <code>zloop</code>).
nphil			number of ϕ positions for the loop calculations.
nphse			number of ϕ positions for the eddy current plots.
nph			number of such contours in ϕ .
mx, mz			Contour grid for χ .
xofsl			offset of the loop positions.
aloop			<code>iloop = 1</code> : distance of the loop "dee" from plasma. <code>iloop = 3</code> : distance of the loops from the plasma. <code>iloop = 4</code> : distance of the loops from the wall.
bloop			elongation of the loop "dee".
dloop			triangularity of the loop "dee".
rloop			<i>not relevant</i> .
ntloop			number of loop positions distributed along the shell.
deloop			delta fraction of <code>plrad</code> to calculate magnetic field from the derivative of χ .

6.7 sprk

Variable	Paper equiv.	Value or type	Definition
Spark and Feedback type variables. Under Development			

For the plasma-independent equivalent, substitute the following definitions. The rest have the same meaning as above but not scaled to the plasma.

Variable	Paper equiv.	Value or type	Definition
cw a	c_w a		Center of the shell. Inner radius of the shell.